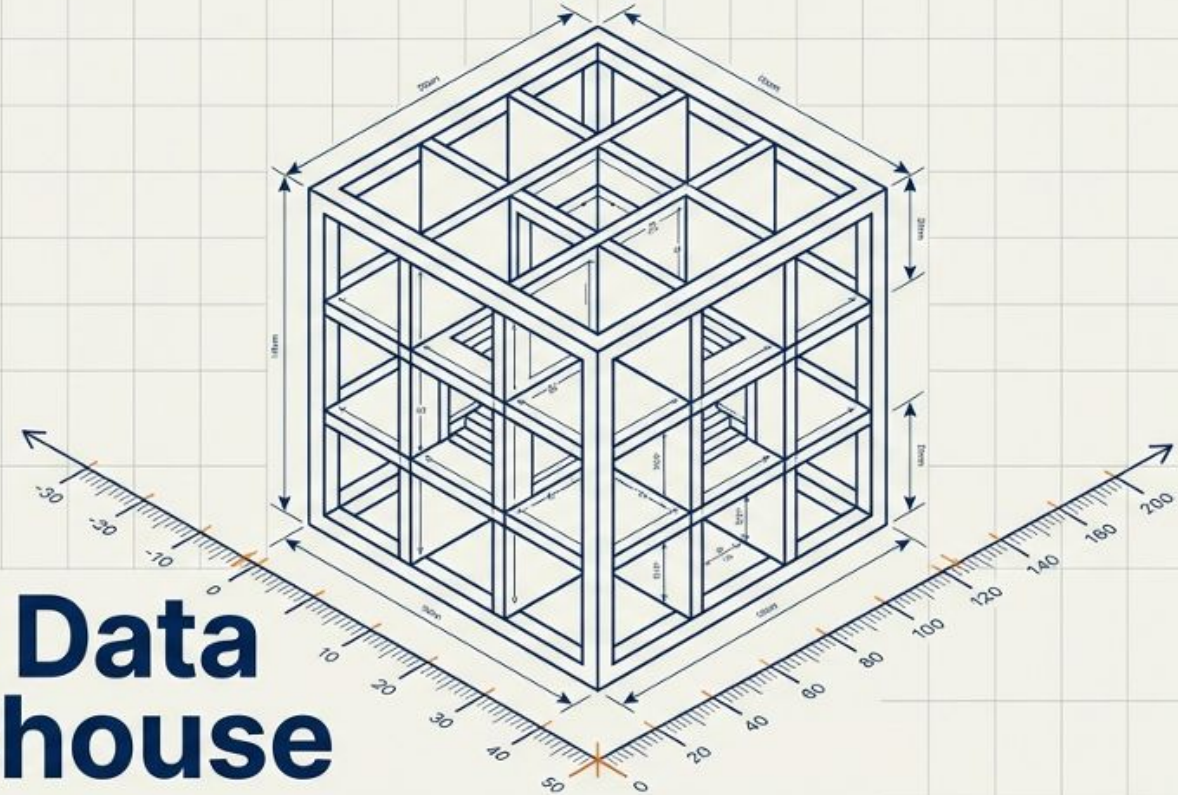


Olist Data Lakehouse

Architectural Blueprint & Business Intelligence Engine

Olist E-Commerce Data Lakehouse

Project Oveview

📦 Strategic E-Commerce Analytics via Google Cloud BigQuery & Modern Data Stack

Transforms fragmented Brazilian e-commerce transaction logs into a high-performance **Star Schema** for executive reporting. By automating the end-to-end flow from raw data to BI, we eliminated 100% of manual data preparation, providing a "Single Source of Truth" for Olist's 9+ disparate datasets.

🛠️ Methodology: The Medallion Architecture

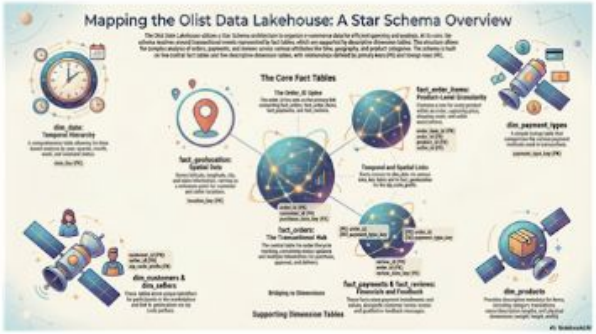
Implemented a modular **ELT (Extract, Load, Transform)** pipeline centered on data reliability and scalability:

Ingestion (Bronze): Automated raw data extraction via **Meltano** into **Google BigQuery**, preserving full data lineage.

Transformation (Silver & Gold): Fact/Dimension models, leveraged **dbt** to normalize schemas, standardize timestamps, and enforce **79+ automated data quality tests**.

Analytics (Rose): Developed denormalized **Data Marts** optimized for low-latency querying and executive-level visualization.

Note: alt+click to navigate the buttons below.



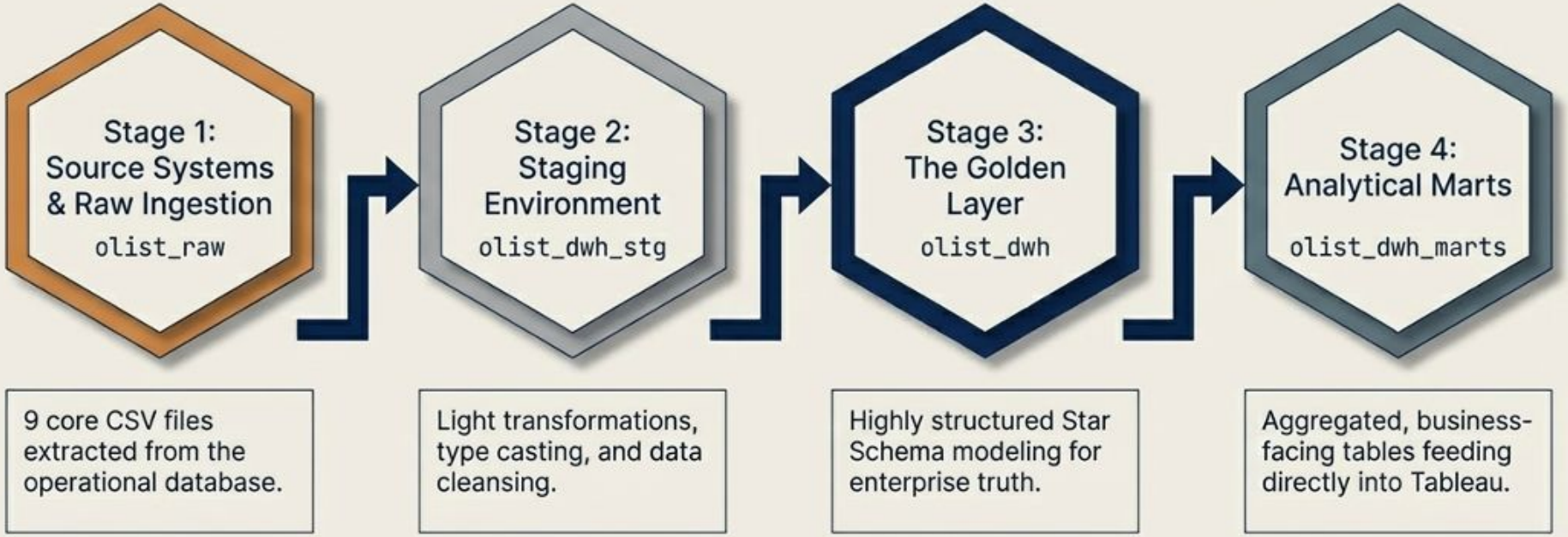
Quality & Trust: Integrated **Great Expectations** for data contract validation and **Tableau** for high-fidelity KPI tracking (e.g., Year-over-Year growth).

🚀 Key Insight: August 2018 Performance

This BI demo illustrates the engineered pipeline from Rose Marts to BI, showing how leadership can quickly move from observing trends to performing root-cause analysis of product categories. MoM revenue fell 4.13% in Aug 2018, despite total revenue rising 20% YoY.

Design & Planning | Meltano Ingestion | dbt Transformation | dbt Test & Great Expt | Rose Marts Preview | BI - Demo

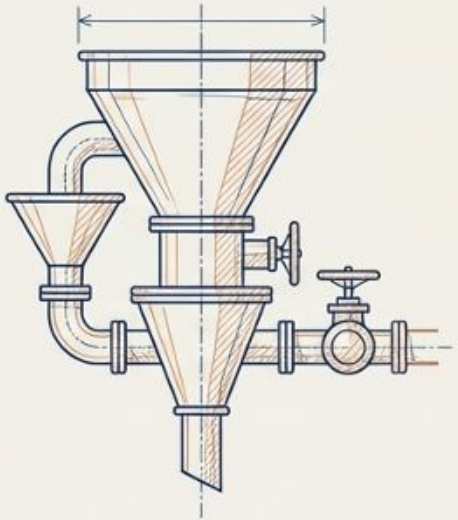
The Modern Data Stack Architecture



Design and Planning

Pipeline Mechanics: Ingestion to Transformation

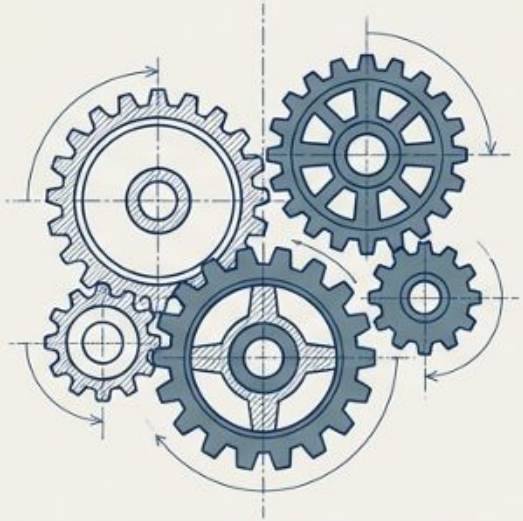
Meltano



Handles **extraction, loading**, and **orchestration**. Ingests 9 heavy operational files directly into the olist_raw layer.

raw_sellers	(174.7MB)
raw_geolocation	(61.27MB)

dbt (Data Build Tool)

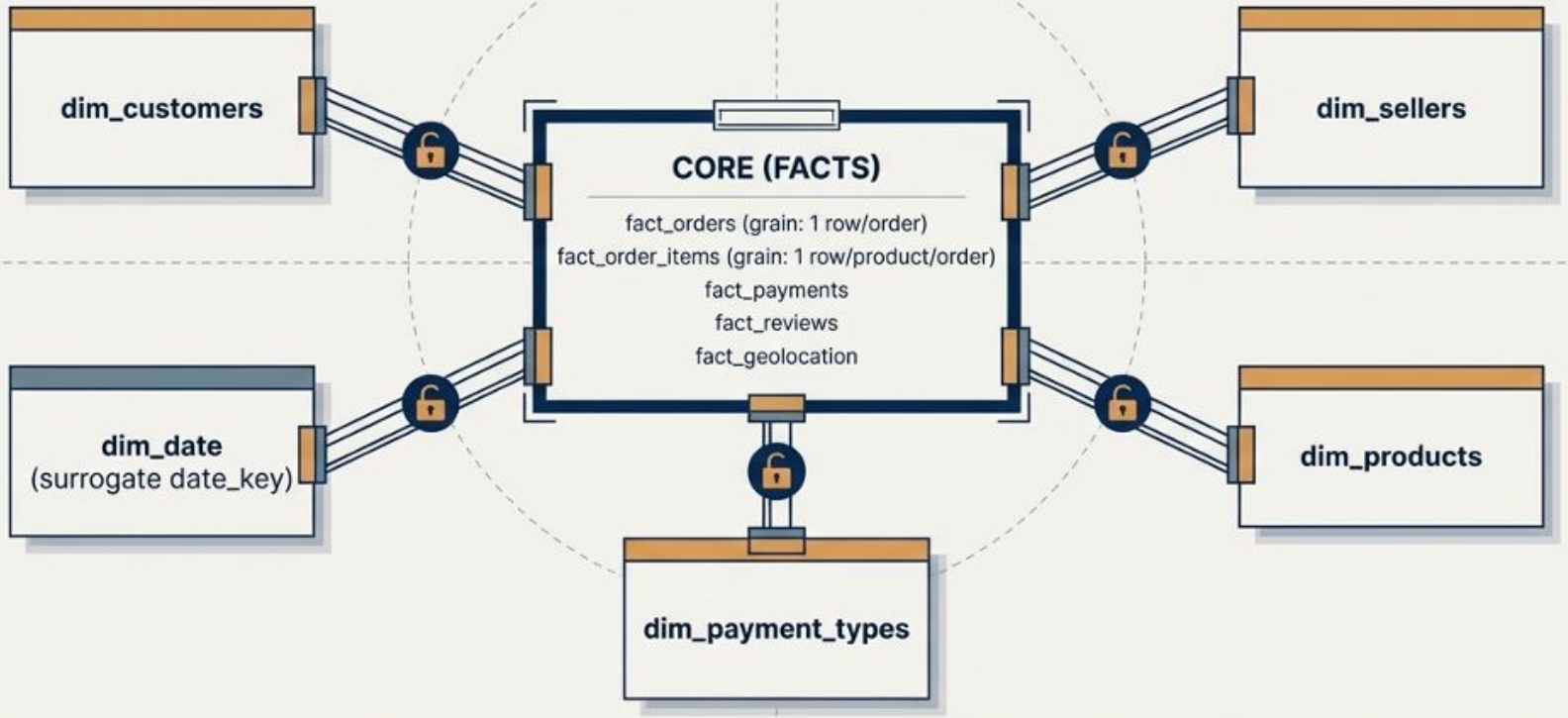


Executes rigorous, **version-controlled SQL transformations.** Standardizes strings, manages surrogate keys, handles safe casting, and tests data integrity from staging up to the analytical marts.

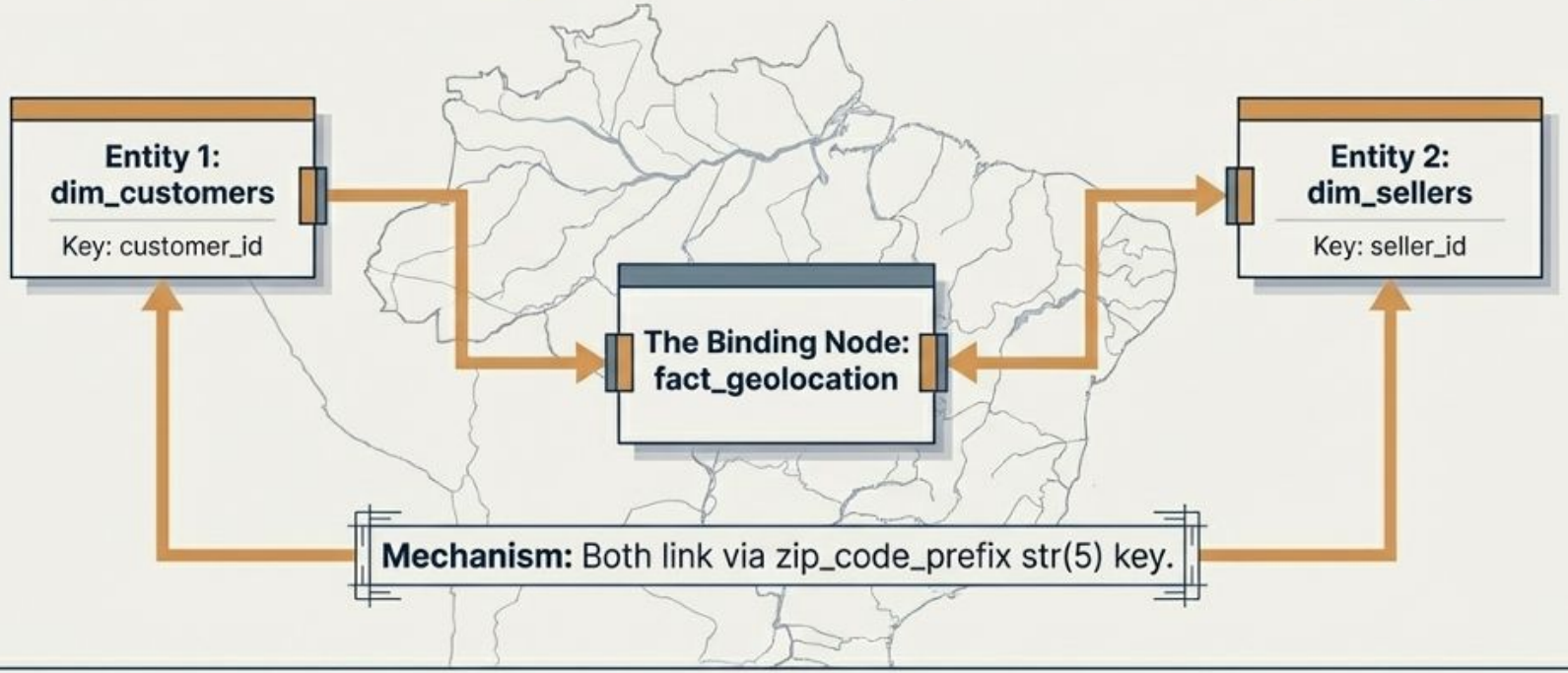
LOWER()
UPPER()

The ‘Golden Layer’ Star Schema

A fully normalized, single source of truth for the Olist enterprise.

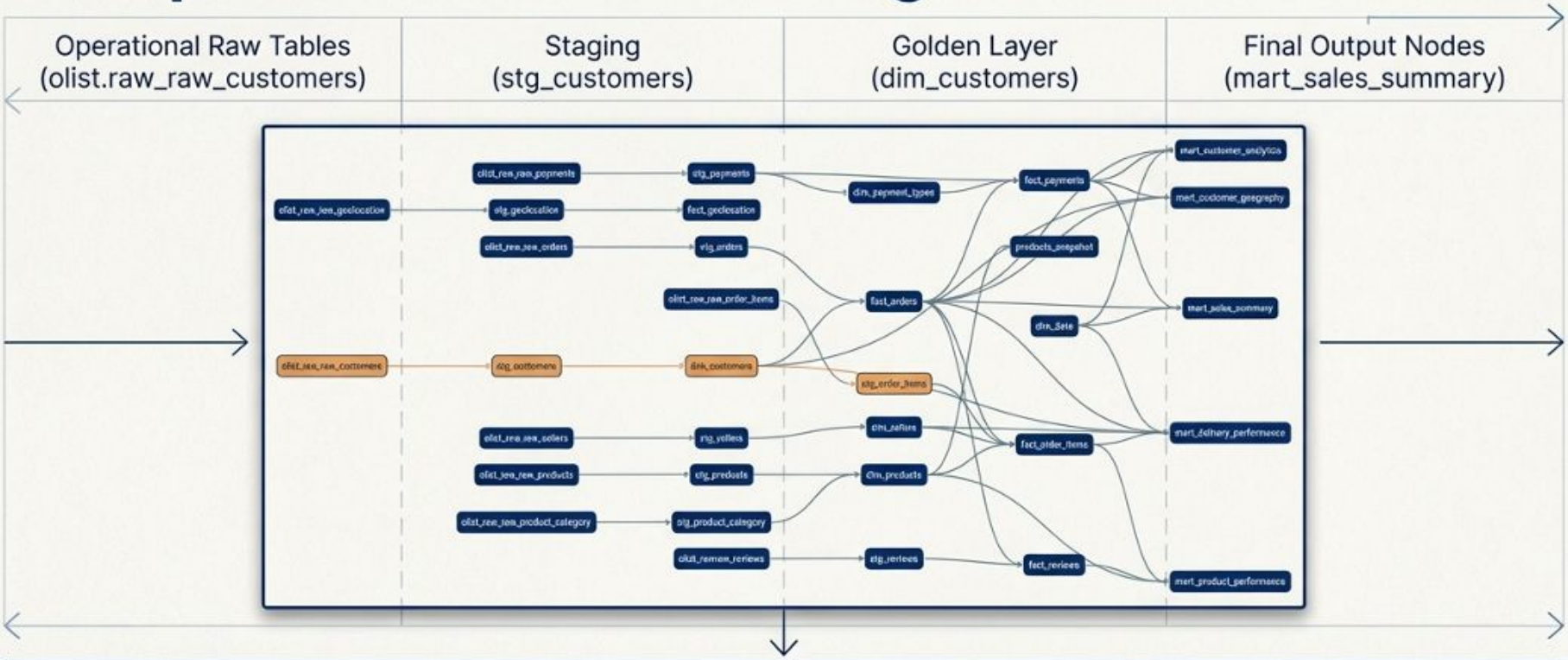


Geospatial Intelligence & Logistics Mapping



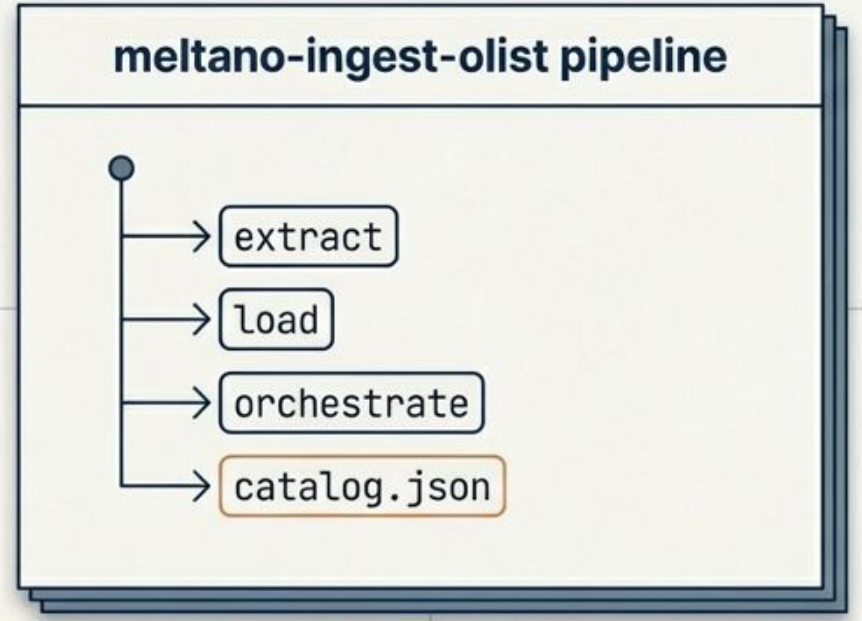
Business Impact: This exact relational join unlocks the ability to map delivery heatmaps, calculate shipping durations, and analyze cross-state commerce between buyers and sellers.

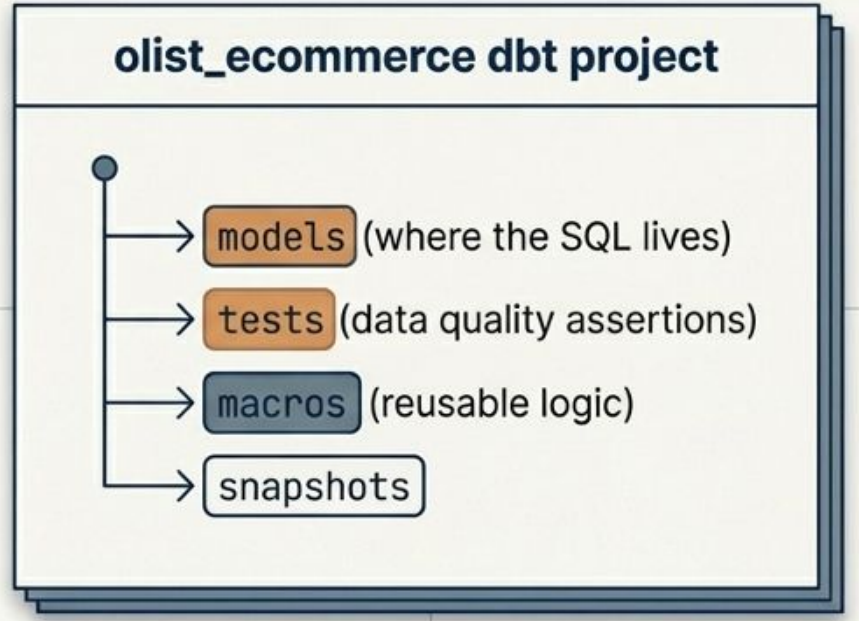
Transparent Data Flow & Lineage



Takeaway: Every metric on the BI dashboard can be traced backward through this lineage graph to its exact operational source, guaranteeing total auditability and isolating pipeline errors instantly.

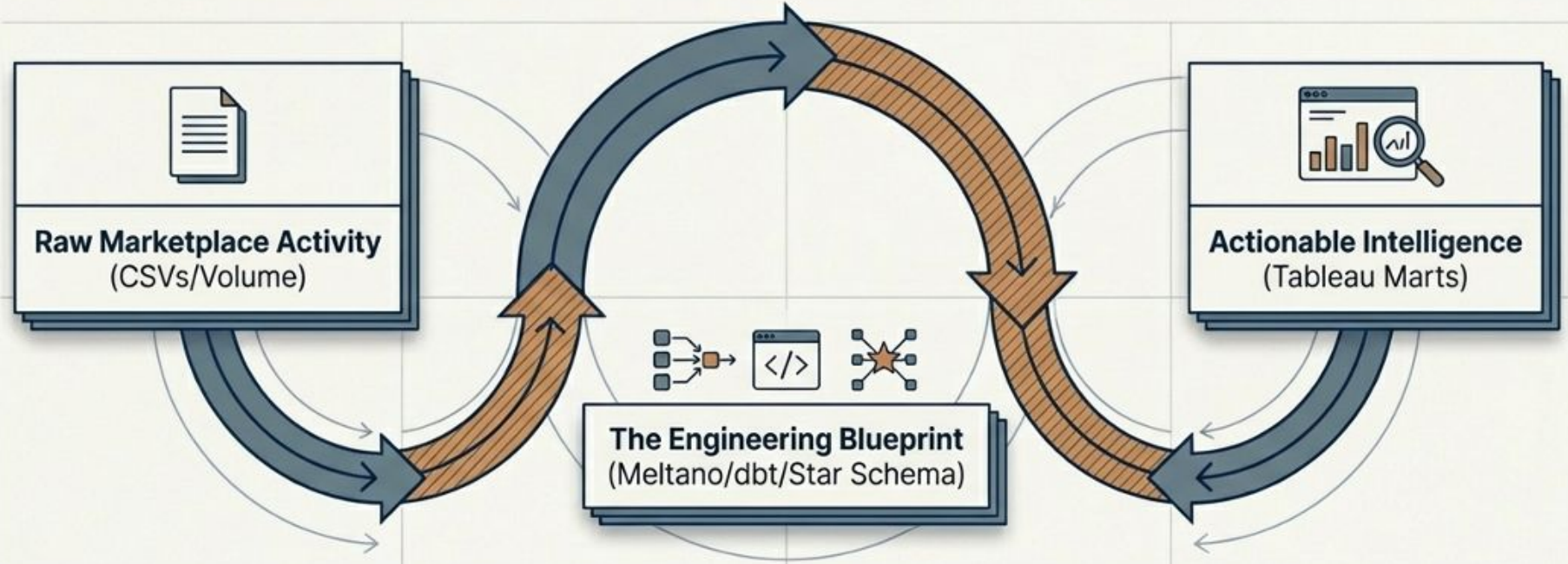
Project Structure & Engineering Rigor





Takeaway: The lakehouse is treated as a software product—fully version-controlled, strictly tested, and consistently logged.

Synthesis: The Lakehouse Value Engine

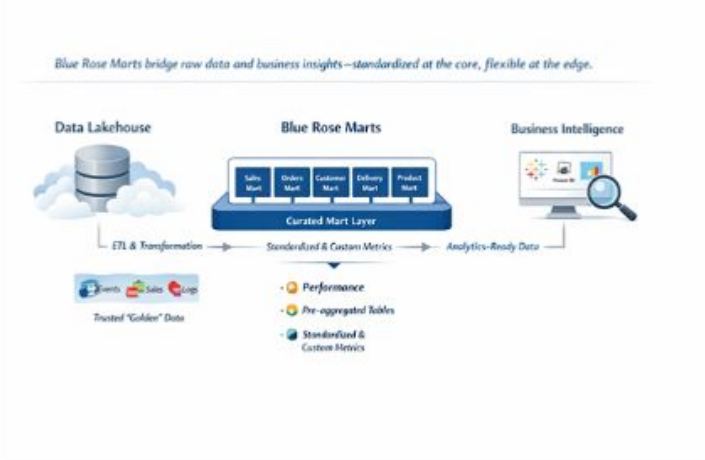


Core Insight: The Olist Lakehouse is not just a storage solution; it is an active translation engine. By applying rigorous software engineering principles to messy operational data, we bridge the gap between backend chaos and front-end clarity, enabling scalable, governed analytics.

Olist E-Commerce Data Lakehouse - Rose Marts

Preview

From Golden Data to Business Impact: The Role of Blue Rose Marts



Note: Alt+Click navigate buttons below.

Why Rose Marts Matter	What Makes Up Rose Marts	How to Leverage - BI Demo

Why Rose Data Marts Matter?

From Trusted 'Golden' Data Lakehouse to Actionable, Business-Ready Insights — Powered by Blue Rose Marts

1. The Challenge (Before)

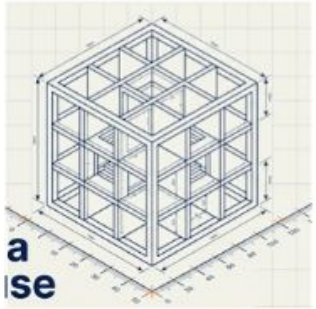
- **Data Silos**: Raw data is fragmented across sources and lacks consistency
- **High Latency**: Complex queries on raw data are slow and resource-intensive
- **Metric Drift**: Different teams report conflicting definitions of key metrics (e.g., "Revenue")

2. The Strategic Value of Blue Rose Marts

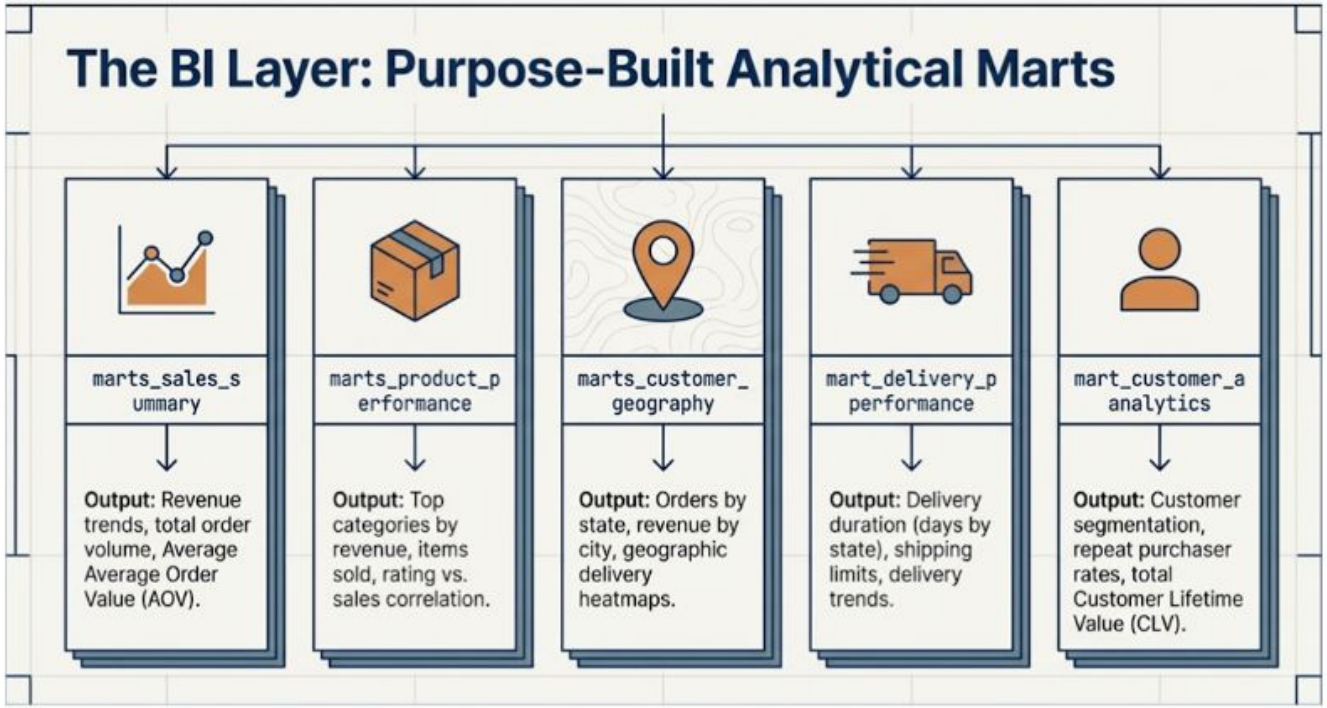
- **Performance**: Pre-aggregated, optimized tables significantly improve dashboard speed (up to 80% faster)
- **Governance**: Role-based access control ensures secure and controlled data usage
- **Consistency**: Standardized KPI definitions create a single source of truth across the organization

3. Key Outcomes — Blue Rose Marts

- **Cost Efficiency**: Reduced compute costs by querying smaller, well-modeled datasets instead of raw data
- **Self-Service BI**: Enables analysts to build reports independently without complex SQL joins
- **Reliability**: Ensures high data quality through automated testing in the transformation layer (e.g., dbt tests)



What Makes up Blue Rose Marts?



Blue Rose Marts represent the **business-ready layer** of the Olist data lakehouse, delivering **essential, standardized metrics** while enabling **flexible customization** to support diverse departmental needs.

How to Leverage Blue Rose Marts?

Executive Sales Overview *(Sample of 5,000 records from Jan 2017–Sept 2018)*

MoM revenue fell 4% in Aug 2018, despite total revenue rising 20% YoY. This demo illustrates the pipeline from Rose Marts to BI, enabling leadership to perform root-cause analysis.

Future enhancement:
city/state filter for drill-down insights.

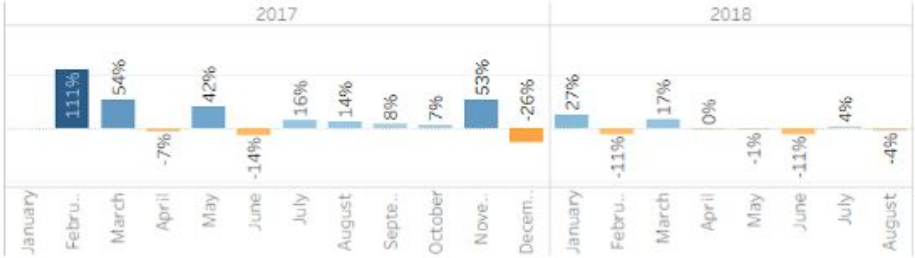
Total Revenue	Revenue MoM %	Total Order Volume	Avg. Order Value
$8.7M	▼ 4.1%	99K	$156
v.s yrs17 (20.1%)	(Aug v.s July 2018)		

Revenue Seasonality (Sales Heatmap)

Dark blue highlights peak sales months.

Revenue MoM Growth %:

Spiked in Nov 2017 (Black Friday), and then trended down to -4% by Aug 2018.



Average Order Value (AOV)

Steady around $156, dripped to $147 in Aug, then surged to $224 in Sept 2018.

Revenue by Day of Week (2018)

Higher spending observed on weekdays compared to weekends.

The Foundation is Set

1

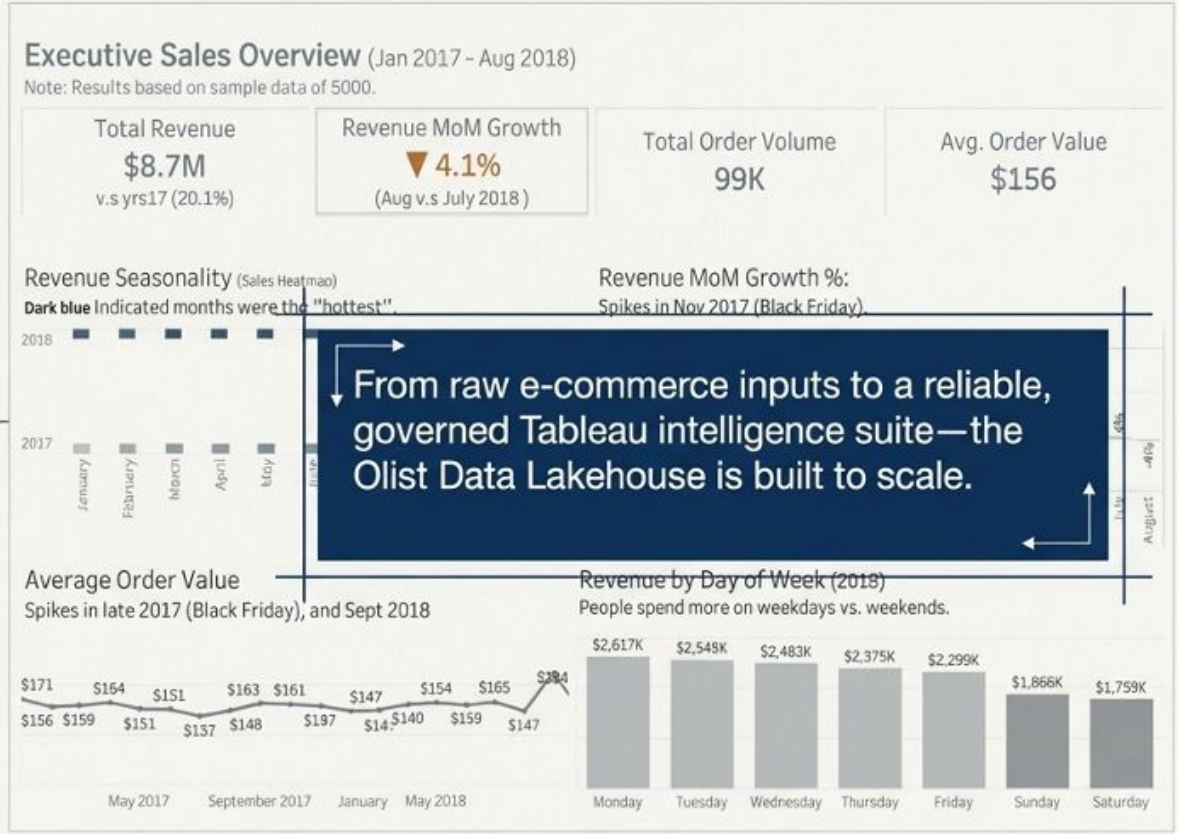
Automated, reliable ingestion pipeline.

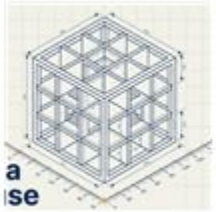
2

Governed, test-driven Star Schema.

3

Five dedicated business-intelligence marts.





Olist E-Commerce Data Lakehouse

Q&A

That concludes my walkthrough of the Olist E-Commerce Data Lakehouse.

I've documented the full ELT logic and dbt test results in my GitHub repository, which you can see from the link:

https://github.com/vivian-sketch58/olist-ecommerce-data-lakehouse.git.

Thank you for your time - I'm now happy to take any questions.